

Software Engineering LAB 3

Amulya P

PES1UG24AM458

Architecture Selection and Justification

Selected Architecture: Layered Architecture

Scenario: Personal Subscription & Recurring Expense Auditor

Reason 1:

Layered Architecture separates the system into different responsibilities such as the User Interface, Transaction Data Manager, Recurring Transaction Analyzer, Subscription Management & Alert Manager, and Expense Database. This makes the system easier to understand and organize.

Reason 2:

Each layer/component has a specific responsibility. For example, the Transaction Data Manager handles transaction data, while the Recurring Transaction Analyzer identifies recurring payments. This makes the system easier to maintain and modify without affecting the entire system.

Security Advantage:

Layered Architecture improves security by keeping database access within the data layer. The User Interface does not directly access the Expense Database, reducing the risk of unauthorized or uncontrolled data access.

Performance Benefit:

Separating transaction management and recurring-transaction analysis allows each part of the system to process its specific task efficiently. This can reduce unnecessary processing and improve the overall response time of the system.

Component Diagram

